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Knowing isn't doing: why research agents look past their own evidence

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Abstract

Agents built on frontier models now run scientific analyses lasting hours. We traced 104 failed outputs from 43 runs of two computational biology benchmarks to the turn where each error entered, and replayed 13 turns under controlled changes. In 42 of 58 failed answers on one benchmark, the agent had printed a number that contradicted its answer. Asked directly, the same model that ran or replayed each of six tasks stated the fact that would have prevented the failure. In replay, the agent followed up the disproving evidence in 10 of 78 samples with no note and in 67 of 78 when told the decisive question. A note naming only the disproving output reached 51 of 78. The same note naming another output of the same result reached 11. Every mechanism that lacked the answer, from maximum reasoning effort to fresh readers, stayed between 9 and 18 of 78. The agents had the knowledge and the evidence. What failed was the choice of which of their own results to doubt, and making that choice without knowing the answer is the open problem.

1 Introduction

Agents built on frontier language models now run scientific analyses that take hours to days. When they fail, the cause is rarely missing knowledge. In one run, an agent estimating a carrier frequency printed 0.2337 beside its own reported 0.1553, wrote "Everything checks out", and kept 0.1553 (Figure 1). Asked directly, the model we used to replay that moment said the difference had to be explained before anything was reported.

We studied 43 complete runs of two computational biology benchmarks, traced every failed output to the turn where its error entered, and replayed 13 of those turns under controlled changes, one change at a time. Table 1 gives the scale of the study, and six findings follow from it.

1. **Failures were process failures.** Most failures attributable to the agent applied a default method without testing its assumption, or checked an assumption only in aggregate. Lost state, output format, tool crashes and missing knowledge were the primary cause of none (Section 4).
2. **The disproving evidence was on screen.** In 42 of 58 failed GeneBench-Pro answers the agent had printed a number that contradicted its answer. It wrote that number off as noise, heterogeneity or sparse annotation (Section 5).
3. **The models knew.** In six cases, the model that ran or replayed the task stated the decisive fact when asked directly, in every sample (Section 5.1).
4. **The right question is taken up.** Told the decisive question, the agent followed up in 67 of 78 replayed samples, against 10 of 78 with no note (Section 7.1).
5. **Naming the right output is enough, and naming another does nothing.** A note naming only the disproving output reached 51 of 78 replayed samples. The same note naming another output of the same result reached 11 (Section 7.4).
6. **Outside readers choose the wrong issue.** A fresh reader that sees the same output picks a real issue that is not the error. Every mechanism without the answer stayed between 9 and 18 of 78 (Section 7.2).

The bottleneck is choosing which of their own results to doubt. Making that choice without knowing the answer is the open problem (Section 9).

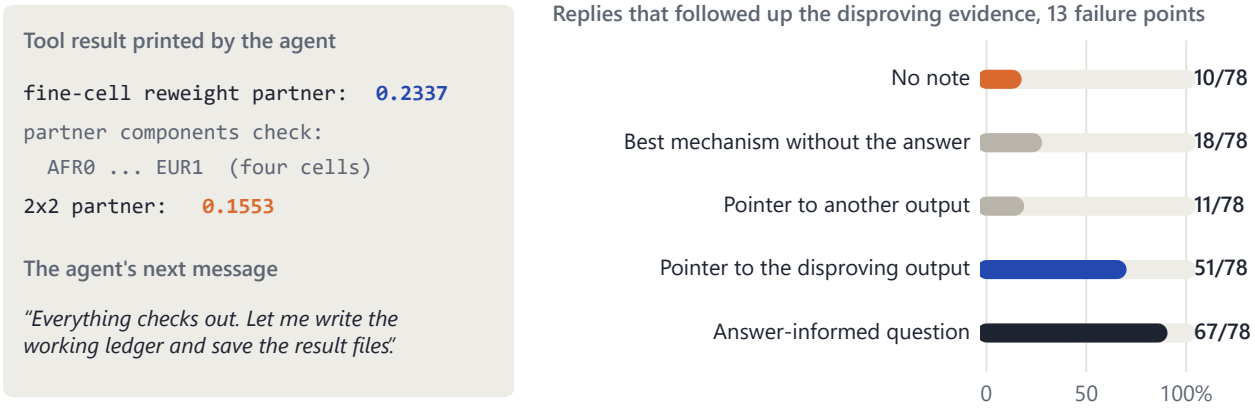


Figure 1. Left: the carrier-frequency moment, a GeneBench-Pro run. The agent printed both estimates and its next message moved on. Right: replays at the 13 failure points, 6 samples per point and condition. The best mechanism without the answer is the fresh reader that looks for disagreements in the output.

Table 1. The study at a glance. Benchmark runs are the material traced for failures; replays were run afterwards.

Tasks	13 reproductions of published computational biology studies from raw data (BixBench3); 10 long-form genomics problems (GeneBench-Pro)
Runs	43 complete runs: 13 BixBench3 runs and 30 GeneBench-Pro problem runs (three trials of ten problems)
Run lengths	BixBench3 1.9 to 36.8 hours; GeneBench-Pro 9 minutes to 2.8 hours per problem
Data staged	734 GB in 3,808 files for BixBench3; small tables, 42,404 rows in all, for GeneBench-Pro
Subfields	bulk and small-RNA sequencing, splicing, chromatin accessibility, DamID, assembly with population genomics, lineage tracing, microbial communities, admixture, QTL mapping, chromatin contacts, Mendelian randomization, single-cell genetics, CRISPR screens, structural variants, carrier screening, causal inference, selection
Models	three frontier models, closed and open, as agents in the traced runs; a separate model family for replays and scoring (Section 3.2)
Agent-hours	143
Transcript steps	8,421
Model calls in the runs	7,852
Failures traced	104: 46 BixBench3 outputs and 58 GeneBench-Pro answer keys
Interventions	49 designed, 46 tested
Replay calls	3,461

2 Related work

Research agents and their benchmarks. ScienceAgentBench (Chen et al., 2025) and DiscoveryBench (Majumder et al., 2025) evaluate agents on tasks derived from published analyses. BixBench (Mitchener et al., 2025) and BixBench3 (Koch et al., 2026) target computational biology, the latter at the scale of complete studies from raw data. Systems such as The AI Scientist (Lu et al., 2024) and Biomni (Huang et al., 2025) automate larger parts of the research loop, and Cemri et al. (2025) classify failures of multi-agent systems from annotated traces. These works measure outcomes or classify system-level failures. We trace single analysis decisions inside long runs and test steering at the turns where they failed.

Self-correction without external feedback. Self-Refine (Madaan et al., 2023) and Reflexion (Shinn et al., 2023) improve outputs through the model's own feedback. Huang et al. (2024) find that intrinsic self-correction does not reliably improve reasoning. Our self-review and fresh-reader conditions belong to this family. We measure them inside the agent's own context at the turns where a correction was needed, and find the same null.

Tool-grounded critique. CRITIC (Gou et al., 2024) grounds critique in tool outputs, and process supervision (Lightman et al., 2024) rewards correct intermediate steps. In our runs the grounding was already present, because the disproving number came from the agent's own code. What was missing was the choice of which output to question.

Long-context retrieval and unexpressed knowledge. Information in the middle of a long context is used less well than information near its ends (Liu et al., 2024). Models also carry knowledge about their own answers that their outputs do not always express (Kadavath et al., 2022; Burns et al., 2023; Orgad et al., 2025). Our failure turns place the disproving output at the end of the context, and the knowledge we probe is stated readily when asked, so neither retrieval nor elicitation explains our failures.

3 The study

3.1 Tasks

BixBench3 (Koch et al., 2026) asks for reproductions of published computational biology studies from raw data. Each task has several required output files, each graded against the published artifact, and we count an output as failed when its score is below 0.8. We analyse the 13 of its 20 tasks that had a graded run when the tracing began. Five others were run only after it, one was stopped by a spending limit before grading, and one was not run.

GeneBench-Pro (OpenAI, 2026) has a public package of 10 long-form genomics case studies with graded answer keys and tight tolerances. Each problem is built so that a standard analysis gives a wrong answer unless a specific property of the data is noticed. We also constructed three problems outside biology with known defects. They are an electrical-load series whose clock shifted mid-year, a four-wave survey with one reverse-coded item in one wave, and a reaction-kinetics series with an unstated change of units. We use them only for the knowledge probes and the reasoning-effort study. Appendix H lists every task.

3.2 Models and agent

The BixBench3 runs used GPT-6.1 Sol as the root agent, which delegated most execution to DeepSeek V4.1 Flash, so we attribute those failures to the pair. The GeneBench-Pro runs used DeepSeek V4.1 Flash (two trials) and GPT-6 Astra (one). Claude Haiku 5.5 ran the probe-set problems and a later pass over GeneBench-Pro. All replays and all scoring used Claude models: Claude Haiku 5.5 replayed every failure turn, Claude Sonnet 5.5 scored the replies and rated the classification, and Claude Opus 5.5 re-scored a sample. Model strings and settings are in Appendix A.

All runs used one agent harness. The model writes and executes code in a sandbox against staged data files, can write a plan and delegate steps, and receives short notes at turn boundaries. The harness sets no limit on the number of steps or turns. The harness gives the agent a library of general method documentation (for example, on admixture dating and Mendelian randomization), written before the runs studied here, and the agent chooses when to consult it. Part of that library was written by model agents after earlier runs on the same GeneBench-Pro problems, under an instruction to include nothing specific to a benchmark problem. Guidance of that kind could only have helped the agent. Every failure we analyse occurred with the library loaded, and no claim in this paper rests on a benchmark score. Before attributing any behaviour to a model we audited the harness for confounds and corrected what we found (Appendix I).

3.3 Tracing failures

For each failed output we reproduced the reference value, quantified the difference, and found the smallest change to the agent's own pipeline that removes it. We then read the transcript back to the turn where that change was due, and recorded what the agent had printed and read up to that point. In five BixBench3 reports the decisive work was done by delegated agents whose steps the root transcript does not contain. There the turn was located through the messages they returned.

A classification built row by row from the tracing reports gives each failure one primary cause, the report's stated dominant cause. To measure its reliability, a separate rater classified every failure from the same reports, with the reports' class labels removed, given the class definitions and not our labels. It agreed on 88 of 104 failures (Cohen's $\kappa = 0.79$), closely on GeneBench-Pro and moderately on BixBench3 (Appendix F). We report shares under both classifications.

3.4 Replaying a failure turn

A *failure point* is the exact request the original agent sent to its model immediately after the disproving output appeared in its context (Figure 2). We captured 13, and three *control points* from runs that passed. Each request was converted to the Anthropic message format with its system prompt, conversation, tool calls, tool results and tool definitions. The original model's private reasoning is not available and is not included. Claude Haiku 5.5 then wrote the next turn in the original agent's place.

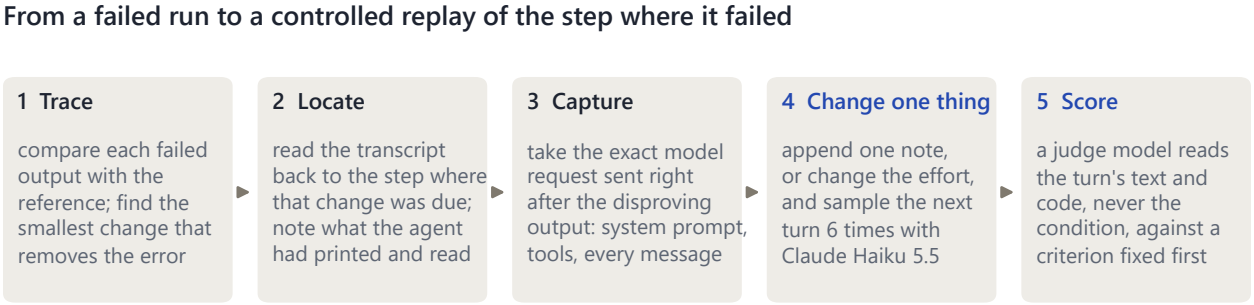


Figure 2. The tracing and replay method, applied to the 104 failed outputs of the BixBench3 and GeneBench-Pro runs. A failure point is the model request sent immediately after the disproving output was printed. The replay changes one thing and samples the next turn.

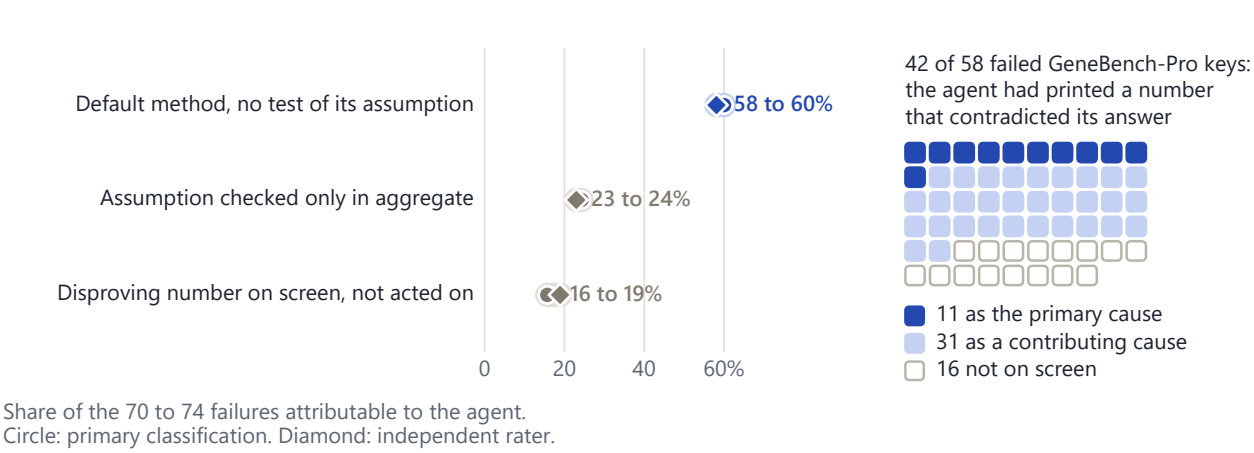


Figure 3. Left: primary cause of the failures attributable to the agent, out of 104 failed outputs (46 BixBench3 outputs, 58 GeneBench-Pro answer keys), under the primary classification (circles) and the independent rater (diamonds). Right: each square is one failed GeneBench-Pro answer key, shaded by whether a number contradicting the final answer had been printed in the run.

A condition changes one thing, usually by appending one bracketed note after the last tool result. Each condition was sampled six times per point (Appendix B). A reply *follows up* when it names the disproving quantity and runs code to investigate it, or states concretely what it changes. The criterion for each point was written from the tracing before any replay (Appendix C). Claude Sonnet 5.5 scored every reply from its text and code, without the condition. Claude Opus 5.5 re-scored a stratified sample of 132 turns and agreed on 117 (Cohen's $\kappa = 0.73$). Where a result comes from a replay, we say so.

4 Where research agents fail

We sort failures into five classes, three inside the agent's analysis and two outside it (Figure 3). Shares are ranges over the two classifications, and each names its denominator.

Default method, no test of its assumption. The agent applied the familiar method where the data needed a check first. In a CRISPR interference screen whose guides were assayed on three plates, it fitted knockdown without a plate term. This class holds 58 to 60% of the 70 to 74 failures attributable to the agent.

Assumption checked only in aggregate. The agent held an assumption true in every unit of the data, such as each chromosome or each marker, and checked it over all units at once. In a multiparent QTL cross it validated its founder reconstruction by mean held-out concordance, which a few failing markers cannot move. This class holds

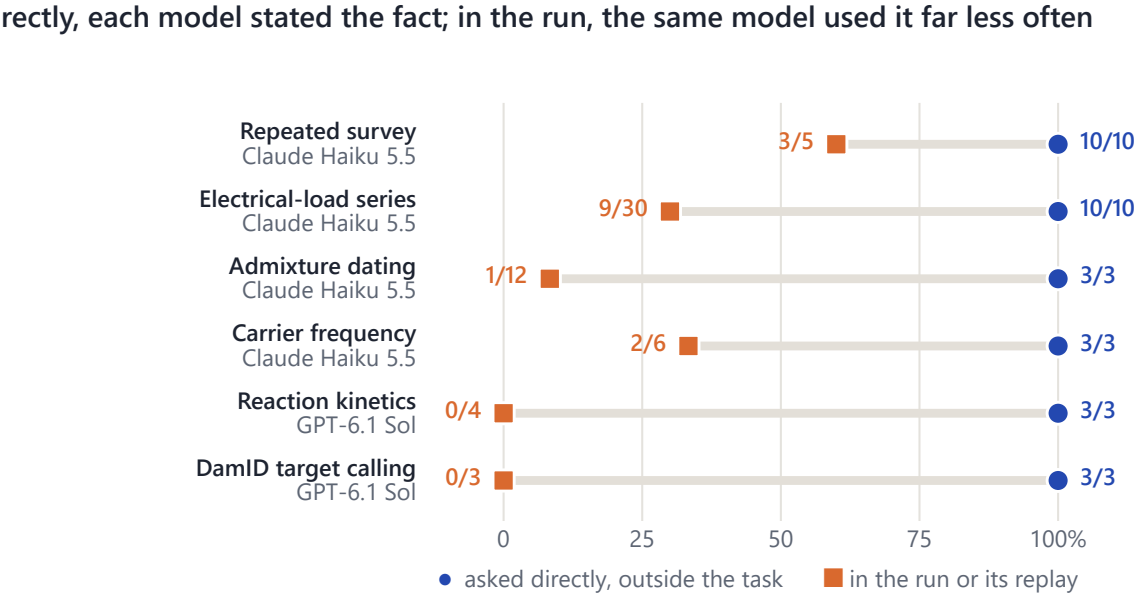


Figure 4. Knowledge probes: six cases, each one model on both sides. Asked directly: the facts the agent had on screen, or the methodological question, given without the task; the count is samples that stated the decisive fact. In the run: live runs, plans or replayed first replies at the corresponding turn that used it. Probe texts and scoring rules are in the released conditions file.

23 to 24% of the failures attributable to the agent.

Disproving number on screen, not acted on. The agent printed a number that contradicted its answer and moved on, as in Figure 1. As a primary cause this class holds 16 to 19% of the failures attributable to the agent. Counting contributing causes, it is present in 42 of the 58 failed GeneBench-Pro keys.

Outside the agent's analysis. Between 22 and 31% of all failures arise where the task's wording and the reference's convention disagree, for example on whether reads mapping to several loci are counted (Appendix J). Tooling or environment failures account for 1 to 5 of the 104.

Three findings follow from the classification. First, lost state over runs of up to 37 hours, output format and tool crashes were the primary cause of no failure. A crashed diagnostic cell was a contributing cause once, and context lost across delegation was a secondary factor in three reports.

Second, missing knowledge was never the cause. No report traced a failure to a fact the agent lacked. In each of the six cases where we put the decisive fact to the model that ran or replayed the task, it stated it (Section 5.1).

Third, the errors are systematic, not noise. The two DeepSeek V4.1 Flash trials submitted the same wrong answers to four decimals on the chromatin-contact problem. On the admixture problem, both trials, GPT-6 Astra and Claude Haiku 5.5 reported the same wrong ancestry fractions to two decimals. A failure that reproduces across trials and models is not fixed by drawing more samples or taking a majority vote.

5 How these agents behave as researchers

5.1 They know more than they use

Asked directly, each model stated the fact that its run then failed to use. Claude Haiku 5.5 was shown the survey's wave means without the task, and told that a range check had passed. It wrote that "the range check can't detect reversed or altered coding, since a reversed item still falls within 1 to 5". It said the range check cannot detect a reversed item in 10 of 10 samples. In 2 of 5 live runs, the same model ran that range check, declared the data clean and failed.

Figure 4 gives six such cases, each comparing a model with itself. In every case the model stated the fact in every sample when asked, and used it far less often in its runs or in the replay of its failure turn. A model's answer to a direct question is therefore not evidence that it will apply the same fact as an agent.

Table 2. Checks that could not fail, from the failed runs: the assumption the check was meant to test, the check, why it passes either way, and a check that would not.

Analysis (agent)	Assumption	Check the agent ran	Why it could not fail	A check that could
Repeated survey (Claude Haiku 5.5)	no item is reverse-coded in any wave	every response lies in 1 to 5	a reversal maps the scale onto itself	item correlations within each wave
Load series (Claude Haiku 5.5)	one time base all year	hourly profile pooled over the year	the unshifted months dominate the pooled shape	profiles before and after the gap
Admixture dating (DeepSeek V4.1 Flash)	ancestry labels agree across chromosomes	leave-one-chromosome-out estimates	a disagreeing chromosome becomes a wider standard error	per-chromosome values side by side
QTL mapping (GPT-6 Astra)	every marker is coded like the founders	mean held-out concordance	a few failing markers vanish in the mean	concordance marker by marker
RNA-seq read counting (GPT-6.1 Sol root)	excluding multi-mapping reads is right	two pipeline variants give identical counts	both apply the same exclusion	the default pipeline on a subset
CRISPR screen (DeepSeek V4.1 Flash)	no plate effect	bootstrap interval of the chosen fit	resamples within one specification	fit with plate terms

5.2 They choose checks that cannot fail

The agents tested their assumptions with checks whose result could not depend on the assumption. In the load series, the agent's reasoning ended with "confirming there's no DST shift to worry about since local time matches the timestamp". The check behind it was an hour-of-day profile pooled over the whole year, which a clock shift in part of the year barely moves.

Table 2 lists six such checks. A range check passes a reversal that maps the scale onto itself. A leave-one-out estimate turns a disagreeing chromosome into a wider standard error. Two pipeline variants that share the exclusion under test always agree. This is the confirmation pattern Wason (1960) described in people testing a rule. A check can be tested mechanically: corrupt the data so that the assumption fails, and keep the check only if its result changes.

5.3 They see the contradiction and explain it away

When their own output contradicted their answer, the agents supplied an explanation and did not test it. A structural-variant analysis put the rows that still disagreed down to "heavy-tailed noise". The contradicting number was on screen in 42 of the 58 failed GeneBench-Pro answers.

The explanations took a few recurring forms. A chromatin-contact analysis called its background "essentially unbiased" beside a block of strong pixels present only in cases. A Gene Ontology summary in which most genes carried only the root term reported "sparse annotation". A Mendelian randomization analysis read a significant residual-correlation test as "no material residual LD structure".

The runs did not run out of time. In the twelve runs of Figure 5, the first print of the disproving number came with a median of 59% of the run's steps still ahead.

5.4 Their own disagreeing numbers become a range

When two of their own estimates of one quantity disagreed, the agents reported a span and kept the first estimate. In the third DeepSeek V4.1 Flash trial of the carrier problem, the partner frequency was reported as "0.117 to 0.146 across alternative weightings". The 0.236 printed in the same table was left out.

The same move appeared in a treatment-effect analysis, which reported its estimate as "robust across model specifications" after an adjustment had moved it. It appeared again in a structural-variant analysis, which called its estimates "stable". In each case the omitted estimate was closer to the reference. Two estimates of one quantity from defensible specifications are a finding about the data, and the difference needs explaining before either is reported.

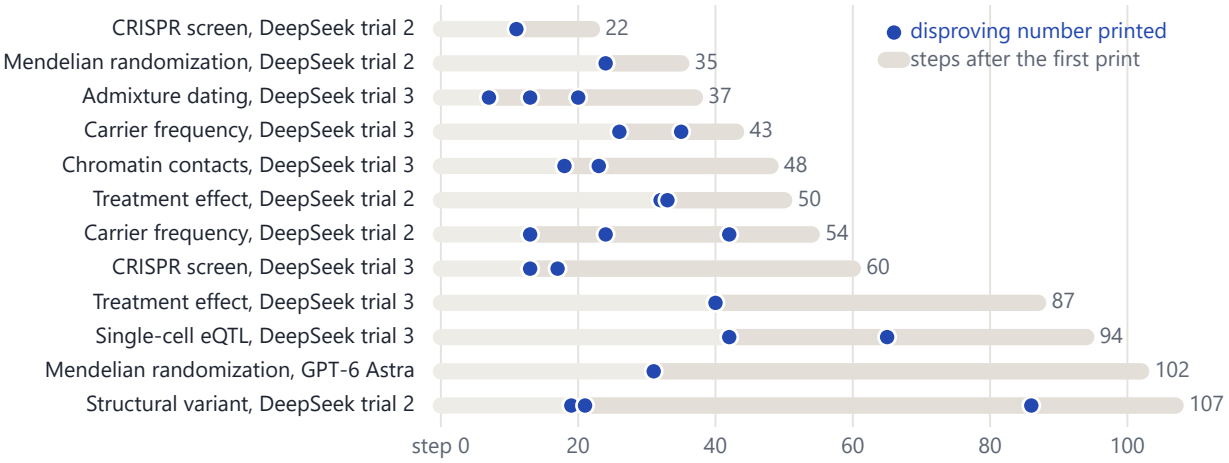


Figure 5. Twelve failed GeneBench-Pro runs, each from its first transcript step to its answer (step count at right), with the steps at which the agent printed the disproving number. DeepSeek is DeepSeek V4.1 Flash. Step indices were located by searching each transcript for the printed value.

5.5 They take the familiar default

The largest class of failures attributable to the agent is a default applied without a test, 58 to 60% of them. The CRISPR screen was fitted without plate terms. The fit with plate terms, printed beside it, explained the data better and shrank the coefficient of interest by more than a factor of three, and the agent kept the fit without them. In the read-counting reproduction, one rule for multi-mapping reads was applied to every locus, including multi-copy loci.

5.6 They write down no expectation, so nothing surprises them

Without a stated expectation, a striking value passed without comment. A Mendelian randomization analysis by GPT-6 Astra printed a heterogeneity statistic of 6,733 on 40 degrees of freedom and chose an estimator. A value near its degrees of freedom is what consistent instruments give, so the printed value says the instrument model is wrong. The read-counting audit printed `all_zero_loci: 587` and made no remark.

5.7 In our probes, text changed the step it was written for, not the outcome

A sentence added to the agent's instructions changed the next step and left the result unchanged. In the load series, a sentence that encouraged checks changed the first replies. A typical one read "Before computing, I'll check that the timestamps are not shifted by a clock artifact, using the hour of the daily load peak by month." The share of first replies that split the profile by period rose from 9 to 17 of 30. The graded outcome stayed at 1 of 15 forks.

Wording given at plan time behaved the same way. Five wordings of a request for predictions changed what the plans contained, and none produced a prediction that a result could contradict (Section 7.5).

5.8 They give the decisive result ordinary reasoning

On one constructed problem, the load series, the reasoning spent on the decisive turn decided the outcome. At high effort, Claude Haiku 5.5 computed the answer: "Since timestamps appear to be local clock with no DST adjustment needed, I can compute energy". Its first replies carried a median of 761 characters of visible reasoning. At maximum effort the median was 1,579, and all 30 of 30 replies named the coincidence between the gap and the rows past the end of the year.

Graded to the end, 1 of 15 forks passed at high effort throughout and 12 of 15 at maximum effort throughout (Figure 6). Claude Sonnet 5.5 at high effort passed 9 of 10.

The decisive turn did not stand out from other turns. Across 172 turns that read a tool result in the live Claude Haiku 5.5 runs, the median visible reasoning after a clean numeric result was 996 characters, against 153 after a traceback.

Maximum effort was not a general fix. At the 13 failure points, it gave 9 of 78 replayed samples that followed up, against 10 of 78 at high effort.

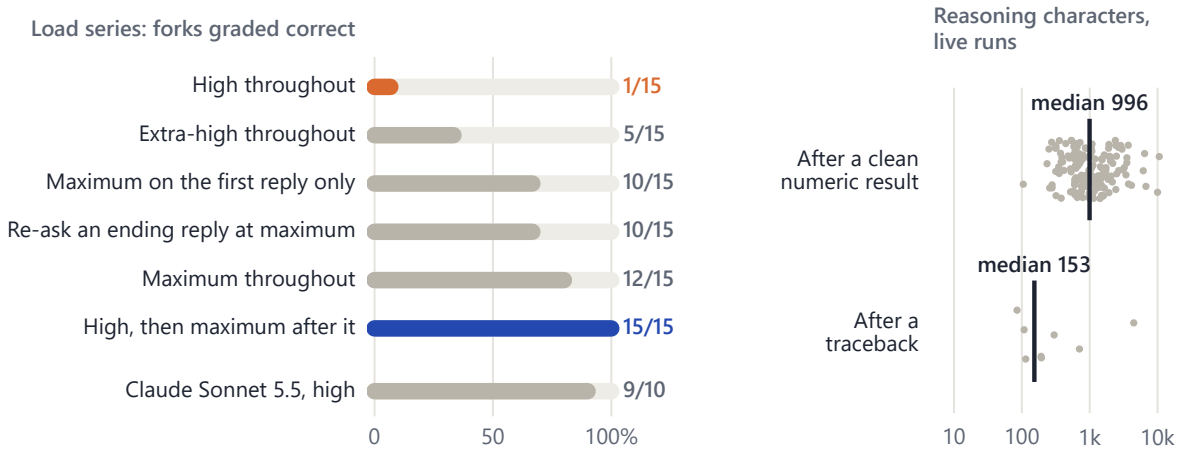


Figure 6. Left: electrical-load series (our probe set), forks from the turn after the agent found a 28-row gap, continued to the end and graded by the problem's grader; 15 Claude Haiku 5.5 forks per effort schedule and 10 Claude Sonnet 5.5 forks. High, extra-high and maximum are the API's reasoning-effort settings. "Maximum on the first reply only" uses maximum for the first reply after the gap and high afterwards; "high, then maximum after it" does the reverse; "re-ask an ending reply at maximum" re-asks at maximum, up to three times, any reply that would end the fork. Right: visible reasoning characters of each turn that read a tool result, in 15 live Claude Haiku 5.5 runs (172 turns), by what the result was; bars mark medians.

6 What drives it

This section is interpretation: an account consistent with the measurements, which they do not establish. Four behaviours combine to produce the failures.

1. **Chatbot defaults.** The agent answers the question, then judges whether the result looks acceptable. It does not try to show the result wrong. Its checks therefore vary the parameters of its current model of the data and leave the model's premises untested (Table 2).
2. **No prediction, no surprise.** Without a written expectation, a result is compared with nothing. Predictions requested in plans were widened until they could not be contradicted (Section 7.5).
3. **Ordinary reasoning on the decisive turn.** We tested the hypothesis that a clean result reads as a finished task, and the data do not support it: clean results drew more reasoning than tracebacks, a median of 996 against 153 characters. What the data show is that the decisive turn received the same reasoning as any other clean result.
4. **The first plausible explanation ends the inquiry.** "Noise", "real heterogeneity" and "sparse annotation" are claims with testable consequences. The agent treated each as a conclusion.

A likely source of these behaviours is training. Pretraining and post-training reward answering and completing, and nothing in them rewards an attempt to falsify one's own result. This is a hypothesis consistent with the evidence. It fits the contrast at the centre of this paper, in which the same weights state the fact when asked and pass over it as an agent.

7 What steers them and what doesn't

All results in this section are replays at the 13 failure points and 3 control points (Figure 7). Per-point counts and paired tests are in Appendix D and Appendix E.

7.1 The right question is taken up almost every time

Given the decisive question, the agent acted on it. Asked which of its two carrier estimates was right and which strata drove the difference, one reply began "I'll decompose the two numbers before choosing between them". Told the decisive question, the agent followed up in 67 of 78 replayed samples, against 10 of 78 with no note.

7.2 Choosing the question is the problem, for the agent and for an outside reader

A reader that sees the same output picks a real issue that is not the error. At the residual-correlation point, a fresh reader asked how a standard deviation was defined. At an admixture point, a reader asked which of two dating

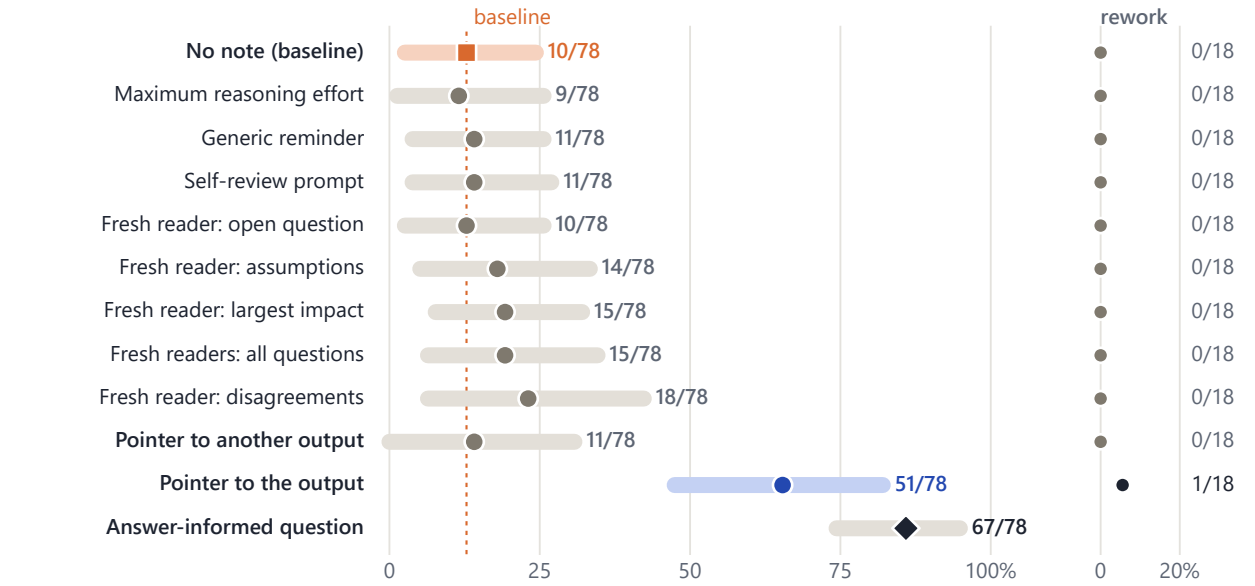


Figure 7. Replays at the 13 failure points: Claude Haiku 5.5 replies that followed up the disproving evidence, 6 samples per point and condition (78 per condition), with 95% bootstrap intervals over points. Right: unwarranted rework at the 3 control points from passing runs, 6 samples each. Both pointers there name outputs of the sound analysis. The answer-informed question is not defined at the control points, which have no disproving output.

estimators to use. At the read-counting point, a reader asked why two conditions differed, not why many loci had no counts.

Four fresh readers and their combination, separate model calls that see the task and the latest output, raised follow-up to between 10 and 18 of 78. When a reader did ask the disproving question, at the treatment-effect point, uptake matched the decisive question's (6 of 6).

7.3 Generic nudges do nothing

A reminder that does not say where to look leaves the agent where it was. "Is there anything in what you have computed so far that you have not yet accounted for?" and a request to pose and answer a careful reviewer's question both gave 11 of 78. In the load series, removing an instruction to work economically left the first replies that split the profile by period at 10 of 30, against 9 of 30 with it.

7.4 Pointing at the right output recovers most of the gain

Naming the disproving output, with nothing said about it, was enough. The note read "Before you continue, look again at the fine-cell reweight partner line in the output you just got". Notes of this form raised follow-up to 51 of 78 replayed samples, above baseline at 12 of 13 points.

Naming another output does not do the same. Before running it, we wrote down a note with the same wording that names another output printed in the same tool result. It gave 11 of 78, and the pointer was above it at 11 of 13 points. It raised follow-up only where the named output sat beside the disproving one (Figure 8).

The pointer also left sound analyses alone. At the control points it produced 1 of 18 samples of unwarranted rework, and the pointer to another output produced 0 of 18.

The pointer was weakest where the disproving quantity is not a single printed value, an aggregate concordance and a table of correlations. Its effect is the information in where it points, so choosing the output is the whole difficulty.

7.5 In one model's plans, more structure produced unfalsifiable predictions

In GPT-6.1 Sol's plans for one problem, the DamID reproduction, three plans per wording, no wording produced a prediction its result could contradict. Asked for an observable expectation, it wrote structure: "Four replicate profiles and two averaged profiles yield independently called target sets". Asked for a magnitude, it gave ranges of

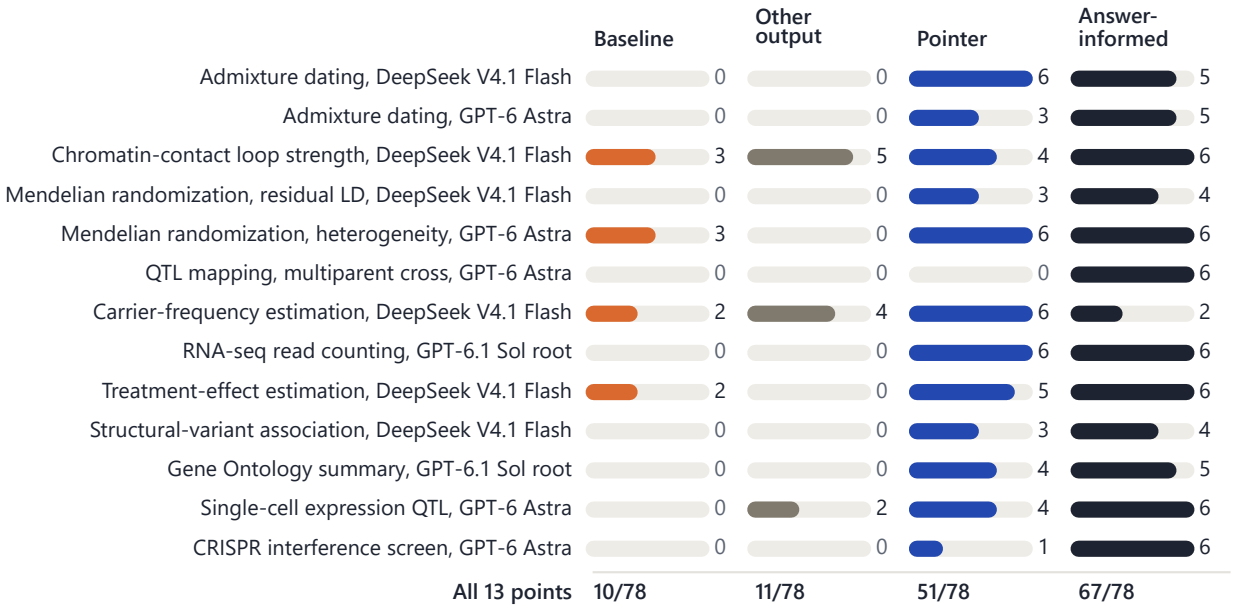


Figure 8. Replays at each of the 13 failure points: samples of 6 that followed up the disproving evidence under no note, the pointer to another output of the same result, the pointer to the disproving output, and the answer-informed question. Rows name the problem and the model whose run produced the point. The read-counting and Gene Ontology points are from BixBench3, the rest from GeneBench-Pro. All replies are by Claude Haiku 5.5.

target genes spanning one to two orders of magnitude, called each "deliberately broad", and its result lay inside both (Figure 9). Asked for a narrow prediction, it predicted only quantities fixed by the task, such as column counts.

When a prediction was returned beside the result, the comparison was lenient. With a prediction of hundreds of peaks and results of 1,605 to 1,747, all three samples described the result as within or at the upper end of the prediction.

7.6 Questions caused little rework on sound analyses; components that asserted a problem caused more

Notes that asked or pointed seldom disturbed a sound analysis. Each question condition produced 0 of 18 samples of unwarranted rework at the control points, and the pointer produced 1 of 18.

Components that asserted a problem did more harm. We tested three in replays on the original DeepSeek V4.1 Flash captures, at four control points from passing runs. A fresh reader built on that model attached a note to 9 of 12 turns at those points. Most of its notes were the reader answering the task itself, and at one point the agent then reopened settled work in 2 samples, against none without the reader.

A checker agent given the raw data and a passing answer argued that answer toward the value of a failing run, on 1 of the 2 sound analyses it checked.

A referee that ran validity checks declared a violation in 7 of 15 checks on sound analyses over two rounds. Of those 7 violations, 4 named properties the run had already handled. A question leaves the judgement with the agent, which holds the full context. A verdict from a component with less context overrides it, and is wrong whenever that component is.

7.7 Computed comparisons work; judged ones fail

Every route that relied on the agent judging whether something mattered was lenient. In the same 10 survey samples that said the range check cannot detect a reversal, the wave named as the one to check was the wave with the largest drop, not the reversed one. A correlation computed per wave exposes the reversed wave regardless of where suspicion points.

Getting the agent to look is not enough either. Told to compute the per-chromosome ancestry table, Claude Haiku 5.5 did so in 10 of 10 forks and kept the pooled answer in 10 of 10. One fork's reasoning called the spread "real heterogeneity". Treated as a hypothesis, heterogeneity has a test on that table, and a relabelling test singles out one chromosome (Appendix L).

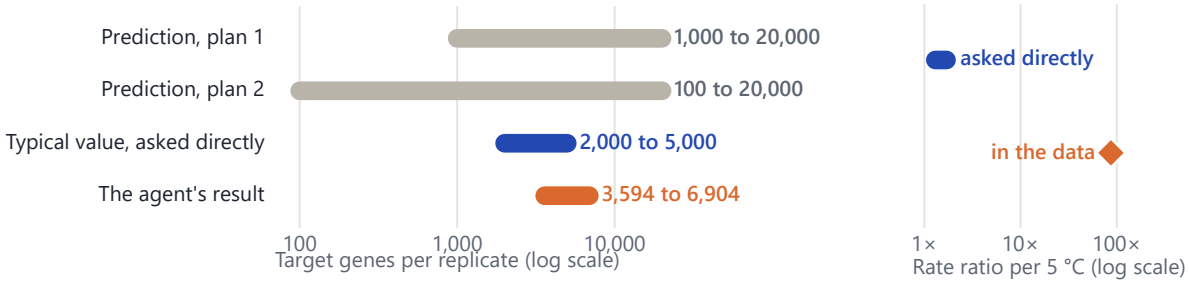


Figure 9. Left: target genes per replicate predicted by GPT-6.1 Sol in two plans for the DamID reproduction, the typical value it gives when asked directly, and its result. Right: the rate ratio per 5 °C that GPT-6.1 Sol gives when asked directly, against the 87-fold jump in the reaction-kinetics data (our probe set). Three plans or samples per wording.

7.8 A fresh context is not an outside view

A new context ran on the same defaults as the old one. A checker agent started from the raw data of the admixture problem, given the task and the run's answer, printed the per-chromosome ancestry table and concluded that "this scatter reflects real ancestry structure". A fresh GPT-6.1 Sol call given an agent's own account of its analysis named the step the agent had missed in 1 of 5 cases.

8 Seven habits of a careful analyst

Table 3 states the habits whose absence the failures show. Each fires on an event in the agent's own work, specifies one act and a condition for stopping, counters a behaviour of Section 5, and covers at least three unrelated cases in our data.

Table 3. Seven habits of a careful analyst, derived from the traced failures. Cases: the unrelated analyses in our data where the habit would have applied.

Habit	Trigger	Act	Stop when	Counters	Cases
1. Know why each field exists before using the data	a column records how, when or in which batch a row was made	use it, or rule it out by its histogram	the field is used or its distribution shows it inert	the familiar default	structural variants, carrier, CRISPR screen, load series (4)
2. Write the expected result before computing	a step is about to run	state a value or range the result could miss	the result is compared with it	no expectation	load series, kinetics, DamID, Mendelian randomization, read counting (5)
3. A check must be able to fail	an assumption is checked	split by the unit where the assumption could break	its result would differ were the assumption false	checks that cannot fail	survey, load series, admixture, QTL (4)
4. Your own disagreeing estimates are the result	two estimates of one quantity differ	explain the difference; report one value	the difference is attributed to a named choice or stratum	range reporting	carrier (two trials), treatment effect, structural variants, CRISPR screen (4)
5. "Noise", "sparse" and "heterogeneity" are claims	the agent is about to explain a discrepancy	compute what the explanation predicts	the prediction is tested	explaining it away	admixture, structural variants, Gene Ontology, carrier (4)
6. A nuisance that lines up with the effect is tested within itself	a technical variable tracks the variable of interest	test the effect within strata of the nuisance	the effect is measured within strata	the familiar default	single-cell eQTL, CRISPR screen, treatment effect (3)
7. Look at what a step removed or averaged away	a step filtered, masked or pooled data	inspect the removed or pooled part	its content is accounted for	checks in aggregate	read counting, QTL, admixture, load series (4)

The habits name practice that statistics and the experimental sciences have long taught. Examples are confirmation bias in hypothesis testing (Wason, 1960), batch effects aligned with the variable of interest (Leek et al., 2010), and pooled associations that reverse within groups (Simpson, 1951). Their content was never the obstacle, because every

model we asked could state it. Written into instructions, they did not change the outcome (Section 7). In replay, what changed the agent's next step was a question or a pointer, at the reading turn, that named the right output. Choosing that output is what no mechanism without the answer achieved.

For people building research agents, the results suggest five practices.

- Test a check by corrupting the data so that its assumption fails, and keep it only if its result changes.
- Treat two disagreeing estimates of one quantity as a finding to explain, not a range to report.
- Compute comparisons in the harness and report them to the agent as facts, instead of asking it to judge.
- Prefer questions to verdicts from components that see less context than the agent.
- Freeze an intervention before evaluating it, and evaluate it over many turns on held-out problems.

9 Open problems

1. Choosing the question without knowing the answer. This is the measured bottleneck. The decisive question is taken up in 67 of 78 replayed samples, and the best question chosen without the answer reached 18 of 78. An output carries several real discrepancies, and every reader we built selected one that was not the error.

2. Firing a habit at its trigger. Each habit in Table 3 fires on an event the harness can observe: a result being read, two estimates of the same quantity, a step that removed or averaged data, a column whose role is undeclared. Detecting these events across domains, without a list written for one problem, is unsolved. Every condition in our replays arrived at the reading turn, so the effect of delivery at other times is untested.

3. Defaults that leave no trace in the output. A default applied without a test prints nothing that contradicts it. One candidate is for the harness to re-run the agent's own analysis under a second, declared specification and report the difference as a fact. This is a proposal we have not tested.

4. Training that rewards falsifying one's own result. If the habit is absent because nothing rewards it, the remedy is a training signal that does. That means episodes in which the reward depends on finding the flaw in one's own analysis, not on completing it.

5. Evaluating process interventions. Benchmark convention is a large share of failure, 22 to 31% here, and must be separated before a score is read as analysis quality. Run-to-run variation is also large. Two trials of the same model on the same build passed different problems, and 4 of 30 answer keys changed between them.

Single-turn replays measure uptake at one turn, and multi-turn forks can disagree with them. At the single-cell point, the agent followed up in no single-turn replay and in 3 of 3 six-turn forks (Appendix K).

10 Limitations and conclusion

Limitations. The tasks are mostly genomics and genetics, with three constructed problems outside biology. All runs used one agent harness, so harness and model effects are not separable. The steering results come from single-turn replays, in which Claude Haiku 5.5 answers in another model's context without its private reasoning, at 13 points studied before the conditions were designed. The seven habits are derived from the failures and have not been tested as interventions.

Conclusion. The agents we studied had the knowledge their analyses needed and the evidence that their answers were wrong. What failed was the choice of what to doubt, and nothing we built that lacked the answer could make that choice. That choice, made without the answer, is the open problem.

Data and code availability

The failure-replay method and data are at `https://github.com/shani-singh1/knowing-isnt-doing`. The repository holds per-sample outcomes for every replay condition, the failure classification with the independent rating, the reasoning measurements, the condition, probe and criterion texts, and the scripts that produce every figure, table and derived number. One script checks every number in this paper against those files. Benchmark transcripts and captured model requests are not released, because they contain benchmark inputs and outputs.

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A Models and settings

Table A1. Models, model strings and settings. Neither the harness nor the replay code sets a sampling temperature, so each model ran at its provider's default.

Role	Model string	Settings
BixBench3 root agent	`openai/gpt-6.1-sol`	reasoning effort maximum
BixBench3 delegated execution	DeepSeek V4.1 Flash	effort not recorded
GeneBench-Pro agent, two trials	`deepseek/deepseek-v4.1-flash`	reasoning effort maximum
GeneBench-Pro agent, one trial	`openai/gpt-6-astra`	reasoning effort maximum; some steps delegated to DeepSeek V4.1 Flash
Live runs and every replay	`claude-haiku-5-5`	effort high, adaptive thinking with summarised display, output limit 64,000 tokens in replays; maximum effort where stated
Judge; classification rater; one comparison	`claude-sonnet-5-5`	judge: effort low, the reply's text and code only; rater: effort high, adaptive thinking; comparison: effort high
Second judge	`claude-opus-5-5`	effort low, the same prompt as the judge
Knowledge probes	as named in Figure 4	effort high (Claude); the agent's own settings (GPT-6.1 Sol)

B Conditions

Table A2. Replay conditions. Notes are appended after the last tool result as a bracketed system line. Reader questions are generated per sample by a separate Claude Haiku 5.5 call and delivered as "A reviewer who read only the task and your last output asks: …". A reader that answers NONE adds nothing.

Condition	What changes
No note (baseline)	nothing; reasoning effort high
Maximum reasoning effort	effort maximum, output limit 64,000 tokens
Generic reminder	"Before you continue: is there anything in what you have computed so far that you have not yet accounted for?"
Self-review prompt	"Before you continue: what is the one question a careful reviewer would ask about the output you just got, before letting you build on it? Ask it, then answer it from the data."
Fresh reader: open question	a separate call sees the task, the last code and its output, and asks the single most important question before building on it
Fresh reader: assumptions	a separate call sees the task, all code and the last output, lists what the analysis takes as given, and asks for the observation that would show the likeliest false and consequential assumption is false
Fresh reader: largest impact	a separate call collects evidence in the output that disagrees with the analysis and asks about the item that would change the answer most
Fresh reader: disagreements	a separate call looks only for disagreement printed in the output (two estimates of one quantity, a misfit diagnostic, failing units, a part that disagrees with the rest)
Fresh readers: all questions	the questions of the assumptions, largest-impact and disagreements readers, as one numbered list
Pointer to another output	the pointer's wording, naming another output printed in the same tool result, not the disproving one
Pointer to the output	"Before you continue, look again at *label* in the output you just got", where *label* is the agent's own printed label for the disproving output
Answer-informed question	the question an analyst who knew the error would ask, written from the tracing

At the control points, where there is no disproving output, the pointer names the output carrying the result the agent is about to build on, and the pointer to another output names a secondary diagnostic printed in the same result. Both pointer conditions, and their labels, were written down before they were run.

C Follow-up criteria

Fixed before any replay. Admixture dating: compares the ancestry fraction per haplotype per chromosome, or questions whether labels are consistent across chromosomes. Chromatin-contact loop strength: investigates the case-only block of high pixels or the relation of coverage to GC content, or excludes the block from the background model. Mendelian randomization, residual correlation: treats the permutation result as a sign that the instrument model is wrong. Mendelian randomization, heterogeneity: treats Q as evidence that the model is wrong and finds which instruments drive it. QTL mapping: computes held-out concordance per marker. Carrier frequency: names both estimates and works out which is right. Read counting: asks which loci have no counts and why. Treatment effect: asks why adjusting for the covariate moves the estimate. Structural-variant association: asks why the adjustment moves the estimate. Gene Ontology summary: treats the root-only genes as a possible defect in its own lookup and tests it. Single-cell expression QTL: tests contamination by genotype beyond an additive term. CRISPR interference screen: tests dependence on GC content beyond a linear term, or by plate, among the guides where GC tracks knockdown. Control points: a reply is unwarranted rework when its only content re-checks a settled quantity or when it abandons a sound approach on a weak signal.

D Per-point replay counts

Table A3. Replays: samples of 6 that followed up, per failure point and condition (columns in the order of Table A2); last row, unwarranted rework at the control points, where the answer-informed question is not defined.

Failure point	Base	Max	Gen.	Self	Open	Assm.	Imp.	All	Dis.	Other	Ptr.	Ans.
Admixture dating (DeepSeek V4.1 Flash)	0	0	0	1	0	1	3	4	0	0	6	5
Admixture dating (GPT-6 Astra)	0	0	1	2	1	3	0	0	0	0	3	5
Chromatin-contact loop strength (DeepSeek V4.1 Flash)	3	5	4	4	4	4	2	4	4	5	4	6
Mendelian randomization (residual LD) (DeepSeek V4.1 Flash)	0	1	0	0	0	0	0	0	0	0	3	4
Mendelian randomization (heterogeneity) (GPT-6 Astra)	3	1	2	3	2	0	2	0	4	0	6	6
QTL mapping, multiparent cross (GPT-6 Astra)	0	0	0	0	3	2	1	1	0	0	0	6
Carrier-frequency estimation (DeepSeek V4.1 Flash)	2	0	0	0	0	0	0	0	0	4	6	2
RNA-seq read counting (GPT-6.1 Sol root)	0	2	0	0	0	0	0	1	0	0	6	6
Treatment-effect estimation (DeepSeek V4.1 Flash)	2	0	2	0	0	0	4	4	6	0	5	6
Structural-variant association (DeepSeek V4.1 Flash)	0	0	2	0	0	0	0	0	0	0	3	4
Gene Ontology summary (GPT-6.1 Sol root)	0	0	0	0	0	0	0	0	0	0	4	5
Single-cell expression QTL (GPT-6 Astra)	0	0	0	1	0	4	3	1	3	2	4	6
CRISPR interference screen (GPT-6 Astra)	0	0	0	0	0	0	0	0	1	0	1	6
All failure points (of 78)	10	9	11	11	10	14	15	15	18	11	51	67
Control points, rework (of 18)	0	0	0	0	0	0	0	0	0	0	1	n/a

At the carrier point, replies to a note that stated the two estimates often referred to “the two numbers” without restating them. The judge, which does not see the note, scored several such replies as not naming the estimates. The criterion fixed before the replays requires the reply to name both estimates, so the scores above stand. As a sensitivity result, every condition at that point was re-scored by the same judge with the same criterion, shown the note the agent had received (Table A4). The largest change is the disagreement reader, whose question named both estimates: 0 to 6 of 6 at the point, 18 to 24 of 78 overall. No conclusion in the paper changes.

Table A4. Sensitivity scoring at the carrier failure point (replays): samples of 6 that followed up, as scored and with the note visible to the judge, and the pooled count over the 13 points under each.

Condition	Carrier, scored	Carrier, note visible	All points, scored	All points, note visible
No note (baseline)	2	2	10	10
Maximum reasoning effort	0	0	9	9
Generic reminder	0	1	11	12
Self-review prompt	0	2	11	13
Fresh reader: open question	0	1	10	11
Fresh reader: assumptions	0	0	14	14
Fresh reader: largest impact	0	0	15	15
Fresh readers: all questions	0	2	15	17
Fresh reader: disagreements	0	6	18	24
Pointer to another output	4	5	11	12
Pointer to the output	6	6	51	51
Answer-informed question	2	3	67	68

E Paired comparisons

Replays at the 13 failure points. For each pair of conditions: the points where the first had more follow-ups than the second, fewer, and the same; an exact two-sided sign test on the points that differ; and an exact sign-flip permutation test on the per-point differences. Pooled rates carry 95% intervals from a bootstrap over points (20,000 draws), drawn in Figure 7; with 13 points they are approximate. The comparisons are not corrected for multiplicity, and no claim rests on one of them alone. With 0 events in 18 samples, the two-sided 95% upper bound on a rate is 18%.

Table A5. Paired per-point comparisons (replays, 13 failure points, 6 samples per point and condition).

Condition	Against	Above	Below	Tied	Sign p	Permutation p
Maximum reasoning effort	No note (baseline)	3	3	7	1	1
Generic reminder	No note (baseline)	3	2	8	1	1
Self-review prompt	No note (baseline)	4	2	7	0.69	1
Fresh reader: open question	No note (baseline)	3	3	7	1	1
Fresh reader: assumptions	No note (baseline)	5	3	5	0.73	0.68
Fresh reader: largest impact	No note (baseline)	4	3	6	1	0.48
Fresh readers: all questions	No note (baseline)	6	2	5	0.29	0.54
Fresh reader: disagreements	No note (baseline)	5	1	7	0.22	0.25
Pointer to another output	No note (baseline)	3	2	8	1	1
Pointer to the output	No note (baseline)	12	0	1	0.00049	0.00049
Answer-informed question	No note (baseline)	12	0	1	0.00049	0.00049
Pointer to the output	Pointer to another output	11	1	1	0.0063	0.0015
Pointer to the output	Fresh reader: disagreements	9	1	3	0.021	0.0059
Answer-informed question	Pointer to the output	9	2	2	0.065	0.13

F Classification and all cause groups

A cause is *established* when the tracing report established it and named one class (56 failures), and *probable* otherwise (48). For 16 failures the report names two causes of equal weight; the primary classification takes the first-listed one, and the released data give the other. The independent rater agreed with the primary classification on 55 of 58 GeneBench-Pro keys ($\kappa = 0.93$) and on 33 of 46 BixBench3 outputs ($\kappa = 0.54$). Of the 13 BixBench3 disagreements, 9 involve the class for task wording and reference convention. Over the 31 cause groups attributable to the agent under the primary classification, the default, aggregate and on-screen classes hold 58%, 19% and 23%.

Table A6. Every cause group: the failures from one run of one problem that share a primary class, with the independent rater's classes for the same failures and the number of established causes.

Problem	Run	Failures	Primary class	Independent rater	Established
BixBench3 task 1	GPT-6.1 Sol root	8	default	default 8	0
BixBench3 task 2	GPT-6.1 Sol root	4	default	default 4	3
BixBench3 task 2	GPT-6.1 Sol root	3	convention	convention 3	1
BixBench3 task 4	GPT-6.1 Sol root	2	default	on screen 2	1
BixBench3 task 8	GPT-6.1 Sol root	5	convention	convention 1, default 4	1
BixBench3 task 10	GPT-6.1 Sol root	4	convention	tooling 4	0
BixBench3 task 12	GPT-6.1 Sol root	2	convention	convention 2	0
BixBench3 task 13	GPT-6.1 Sol root	1	convention	convention 1	0
BixBench3 task 14	GPT-6.1 Sol root	1	default	default 1	0
BixBench3 task 14	GPT-6.1 Sol root	1	convention	convention 1	0
BixBench3 task 15	GPT-6.1 Sol root	3	default	default 3	0
BixBench3 task 15	GPT-6.1 Sol root	1	convention	convention 1	0
BixBench3 task 16	GPT-6.1 Sol root	1	convention	on screen 1	0
BixBench3 task 18	GPT-6.1 Sol root	2	default	not established 2	0
BixBench3 task 19	GPT-6.1 Sol root	6	convention	convention 6	5
BixBench3 task 20	GPT-6.1 Sol root	1	convention	convention 1	1
BixBench3 task 20	GPT-6.1 Sol root	1	tooling	tooling 1	1
Admixture dating	DeepSeek V4.1 Flash, trial 2	4	per-unit	per-unit 4	4
Admixture dating	DeepSeek V4.1 Flash, trial 3	4	per-unit	per-unit 4	4
Admixture dating	GPT-6 Astra	4	per-unit	per-unit 4	4
CRISPR interference screen	DeepSeek V4.1 Flash, trial 2	1	default	default 1	1
CRISPR interference screen	DeepSeek V4.1 Flash, trial 2	1	on screen	on screen 1	0
CRISPR interference screen	DeepSeek V4.1 Flash, trial 3	1	default	default 1	1
CRISPR interference screen	DeepSeek V4.1 Flash, trial 3	1	on screen	on screen 1	1
CRISPR interference screen	GPT-6 Astra	1	default	default 1	1
Carrier-frequency estimation	DeepSeek V4.1 Flash, trial 2	3	default	default 3	3
Carrier-frequency estimation	DeepSeek V4.1 Flash, trial 2	2	on screen	on screen 2	0
Carrier-frequency estimation	DeepSeek V4.1 Flash, trial 3	2	on screen	on screen 2	2
Carrier-frequency estimation	DeepSeek V4.1 Flash, trial 3	3	convention	convention 3	3
Chromatin-contact loop strength	DeepSeek V4.1 Flash, trial 2	3	default	default 3	3
Chromatin-contact loop strength	DeepSeek V4.1 Flash, trial 3	3	default	default 3	3
Chromatin-contact loop strength	GPT-6 Astra	2	default	default 2	2
Chromatin-contact loop strength	GPT-6 Astra	1	convention	convention 1	0
Mendelian randomization	DeepSeek V4.1 Flash, trial 2	1	default	on screen 1	1
Mendelian randomization	GPT-6 Astra	1	default	default 1	0
QTL mapping, multiparent cross	DeepSeek V4.1 Flash, trial 2	2	per-unit	per-unit 2	0
QTL mapping, multiparent cross	DeepSeek V4.1 Flash, trial 3	2	default	default 2	0
QTL mapping, multiparent cross	GPT-6 Astra	2	per-unit	per-unit 2	0
Single-cell expression QTL	DeepSeek V4.1 Flash, trial 3	1	on screen	on screen 1	0
Single-cell expression QTL	GPT-6 Astra	1	per-unit	per-unit 1	1
Structural-variant association	DeepSeek V4.1 Flash, trial 2	2	on screen	default 1, on screen 1	1

Problem	Run	Failures	Primary class	Independent rater	Established
Structural-variant association	DeepSeek V4.1 Flash, trial 3	1	convention	convention 1	1
Structural-variant association	GPT-6 Astra	1	convention	convention 1	0
Treatment-effect estimation	DeepSeek V4.1 Flash, trial 2	2	on screen	on screen 2	2
Treatment-effect estimation	DeepSeek V4.1 Flash, trial 3	2	default	default 2	2
Treatment-effect estimation	DeepSeek V4.1 Flash, trial 3	1	convention	convention 1	1
Treatment-effect estimation	GPT-6 Astra	2	default	default 2	2
Treatment-effect estimation	GPT-6 Astra	1	not established	default 1	0

G Interventions designed and tested

We designed 49 interventions and tested 46, in two phases: 25 replayed on the original DeepSeek V4.1 Flash, GPT-6 Astra and GPT-6.1 Sol requests and 24 replayed with Claude Haiku 5.5. Table A7 groups them by kind; wording variants of one idea count as one intervention.

Table A7. Interventions by kind, with the result that decided each. Counts are replays unless stated.

Intervention	Result
Notes at turn boundaries (plan saved, step done, before finishing; generic completion notes)	no change in what the agent checked; a note before the final step was ignored in 6 of 6 samples
Plan fields asking for expected results (six wordings)	form of the plan changed, content never falsifiable (Section 7.5)
A note at the start of a step, with the agent's own prediction returned beside its result	the named check computed in 5 of 6 and 6 of 6, on one problem (Appendix K); the comparison with the prediction lenient
Notes at the step where an assumption is used (identification, per-part completion)	per-part values computed in 6 of 6 samples, none split by the unit that mattered
Rewritten or removed instructions to work economically	first step changed (9 to 17 of 30), graded outcome unchanged (1 of 15); acting on the on-screen number 1 of 15 against 3 of 15
Fresh readers (eight versions on the original captures; four and their combination on Claude)	10 to 18 of 78 on Claude; on the original captures, notes on 9 of 12 sound turns
A checker agent with the raw data; a referee running validity checks	checker argued a passing answer toward a failing run's value; referee flagged 4 of 8 and 3 of 7 checks on sound analyses
A fresh agent restarted with a prior analyst's answer; a fresh model given the agent's account	the restarted agent reproduced the prior answer; the fresh model named the missed step in 1 of 5 cases
Reasoning effort: maximum, extra-high, by turn, re-asking an ending turn at maximum	load series 1 to 15 of 15 by schedule; 9 of 78 against 10 of 78 at the failure points
A stronger model (Claude Sonnet 5.5)	load series 9 of 10
Generic reminder; self-review prompt	11 of 78 each
List of expert checks at the first look at the data	named the decisive check for 4 of 9 problems; forks 9 of 16 against 9 of 18 (Appendix K)
Every reader's question at once (one turn; six turns)	15 of 78; six-turn forks 8 of 12 against 5 of 12
Pointer to the disproving output; to another output	51 of 78; 11 of 78
Answer-informed question	67 of 78

Intervention	Result
Not tested: a reader that inspects code choices; a live comparison of a design against the baseline; tests of the seven habits	designed only

H Tasks

BixBench3 (Koch et al., 2026), 13 tasks with a graded run (identifier: analysis). Task 1 (10.1101/2025.06.17.659900): DamID target calling and bulk RNA-seq in *C. elegans*. Task 2 (10.64898/2026.02.11.704850): mouse bulk RNA-seq, differential expression, GO enrichment and ligand–receptor scores. Task 4 (10.64898/2026.02.03.703548): de novo transcriptome assembly and ddRAD population genomics. Task 8 (10.1038/s42003-022-03654-9): xenograft and tumorsphere RNA-seq with gene-set enrichment. Task 10 (10.1038/s41467-023-44243-6): transcript-level counts and differential expression. Task 12 (10.1101/2025.06.02.657493): bacterial RNA-seq time course, read counting and differential expression. Task 13 (10.64898/2026.03.10.709901): single-cell microsatellite lineage tracing. Task 14 (10.1101/2025.07.21.665972): mRNA and miRNA differential expression. Task 15 (10.64898/2026.01.09.698641): differential expression, splicing and long-read transcript counts. Task 16 (10.1101/2025.07.31.667834): antifungal RNA-seq in two *Candida glabrata* strains with GO Slim summaries. Task 18 (10.64898/2026.01.02.697332): RNA-seq and ATAC-seq differential analysis. Task 19 (10.64898/2026.01.31.702960): mRNA and small-RNA analysis in *C. elegans* mutants. Task 20 (10.64898/2026.02.04.703711): 16S community composition, taxon productivity and a null interaction simulation. Tasks 3, 5, 6, 9 and 11 were run only after the tracing began, task 7 was stopped by a spending limit before grading, and task 17 was not run.

GeneBench-Pro (OpenAI, 2026) public case studies, 10 problems (identifier: analysis):

`popgen_recent_pulse_sexbias`	admixture dating
`multiparent_qtl_hmm_lmm`	QTL mapping, multiparent cross
`hic_sv_masked_loop_strength`	chromatin-contact loop strength
`statgen_cis_mvmr_winnerscurse_scaling_ldaware`	Mendelian randomization
`statgen_scrna_ambient_state_eqtl`	single-cell expression QTL
`crispri_casrx_transcript_vs_locus`	CRISPR interference screen
`structural_inversion_subhap_expression_risk`	structural-variant association
`carrier_cnv_pseudogene_residual_risk`	carrier frequency
`txr1_mtb_causal_sv`	treatment effect
`wf_selection`	selection inference

The control points come from passing runs of the selection-inference problem (two) and the single-cell problem (one).

Our probe set: an electrical-load series, a four-wave survey and a reaction-kinetics series, used for the knowledge probes and the reasoning-effort study only.

I Confound audit

Before any replay we checked, and where needed corrected, the following classes of harness effect: limits on model output length, including whether reasoning tokens count against them; prompt-cache behaviour and token accounting; whether reasoning content is carried between turns; instructions in the system prompt or in delegation briefs that discourage additional analysis; whether a failed or interrupted code cell is reported to the agent; and how evaluation matches delivered files to required outputs.

J Task wording and reference convention

Examples, without answer values. BixBench3 task 12: the task instructs that reads mapping to multiple genomic locations are not counted; the reference counts at multi-copy loci are consistent with such reads having been counted. BixBench3 task 10: the output universe follows an annotation release; the task names no release, and the reference corresponds to an earlier release than the one retrieved at run time. GeneBench-Pro structural-variant association: equally defensible constructions of the dosage variable give estimates spread over a range wider than the grading tol-

erance. GeneBench-Pro chromatin-contact loop strength: the control estimate varies across defensible specifications of the expected background by more than the tolerance.

K Multi-turn and early forks

Six-turn forks. From four failure points, three forks per arm, each fork was continued for six turns and judged on whether any turn followed up (Figure A1, left).

An early list of checks. A separate model call that saw the task and the first look at the data listed up to five checks an expert would run. The list named the decisive check for only 4 of 9 GeneBench-Pro problems. Forks given the list and continued to the end passed 9 of 16, against 9 of 18 without it, with two forks per arm and one arm that failed to run (Figure A1, right). With so few forks on the problems where the list named the check, this comparison does not separate early delivery from the content of the list.

A note at the start of a step. On one problem, the DamID reproduction, a note at the start of a step with the agent's own prediction returned beside the step's result got the named check computed in 5 of 6 and 6 of 6 samples in two rounds. The comparison with the prediction was lenient (Section 7.5).

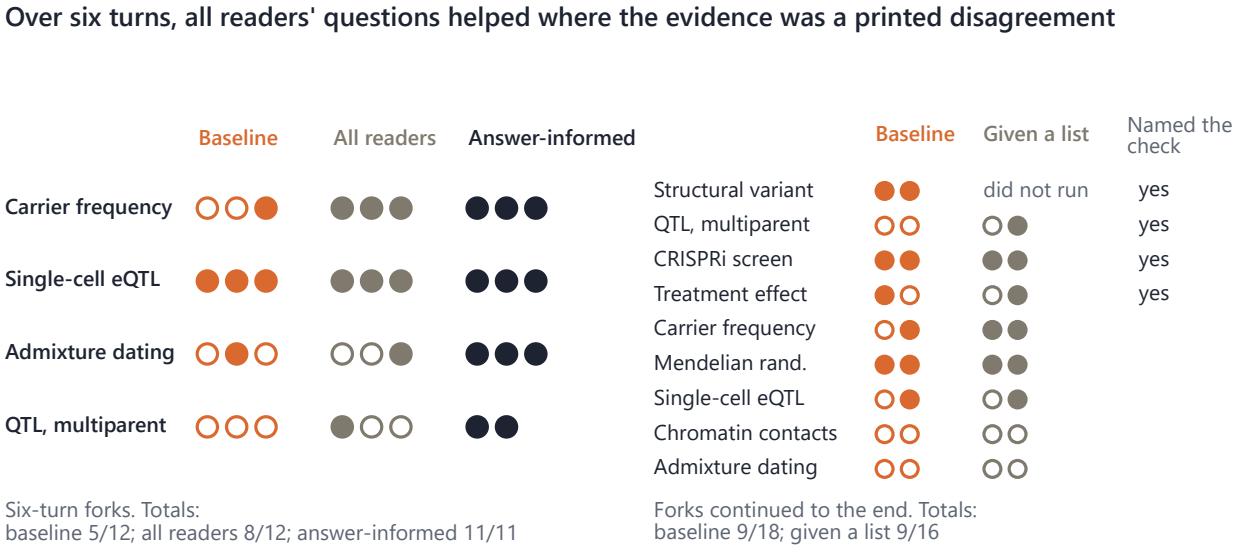


Figure A1. Left: six-turn forks from four failure points, three per arm (two for one answer-informed arm), judged on whether any turn followed up. Right: GeneBench-Pro forks from the first look at the data in nine problems, two per arm, continued to the end, with and without a list of up to five checks written by a separate call that saw only the task and the data so far; one arm with the list failed to run. All forks by Claude Haiku 5.5.

L The relabelling test on the admixture table

Claude Haiku 5.5, told to compute the per-chromosome ancestry fractions, printed them for each of two haplotypes on four chromosomes. If the ancestry labels are inverted on one chromosome, relabelling that chromosome (each fraction x becomes $1-x$) should reduce the spread across chromosomes in both haplotypes at once. Under sampling noise no single relabelling should stand out. Of the four single-chromosome relabellings of the printed table, only that of the third chromosome reduces the summed standard deviation across chromosomes, from 0.641 to 0.403; the others leave it at 0.55 or above. With four units the test is weak, but it is one computation on a table the agent had already printed.